Mixed Traffic Test (Parallel)

2025-10-14-16-42-47

Test Setup Information

Overall Setup Info For all Tests	DUT Model	Test_DUT
	DUT Firmware	NA
	SSID	Test Configured
	Security	Test Configured
	No of Devices	3 (Virtual Clients: 0, Windows: 1, Linux: 0, Mac: 0, Android: 2 ,iOS: 0)
	Test Duration (HH:MM:SS)	00:01:00

Objective

The Candela mixed traffic test is designed to measure the access point performance andstability by running multiple traffic on real clients like Android, Linux, Windows, and IOSconnected to the access point. This test allows the user to choose multiple types of traffic likeclient capacity test, web browser test, video streaming test ping test. Along with theperformance measurements are client connection times, Station 4-Way Handshake time, DHCPtimes, and more. The expected behavior is for the AP to be able to handle all types of traffic onthe several stations (within the limitations of the AP specs) and Make sure all clients can run alltypes of traffic.

Traffic Details

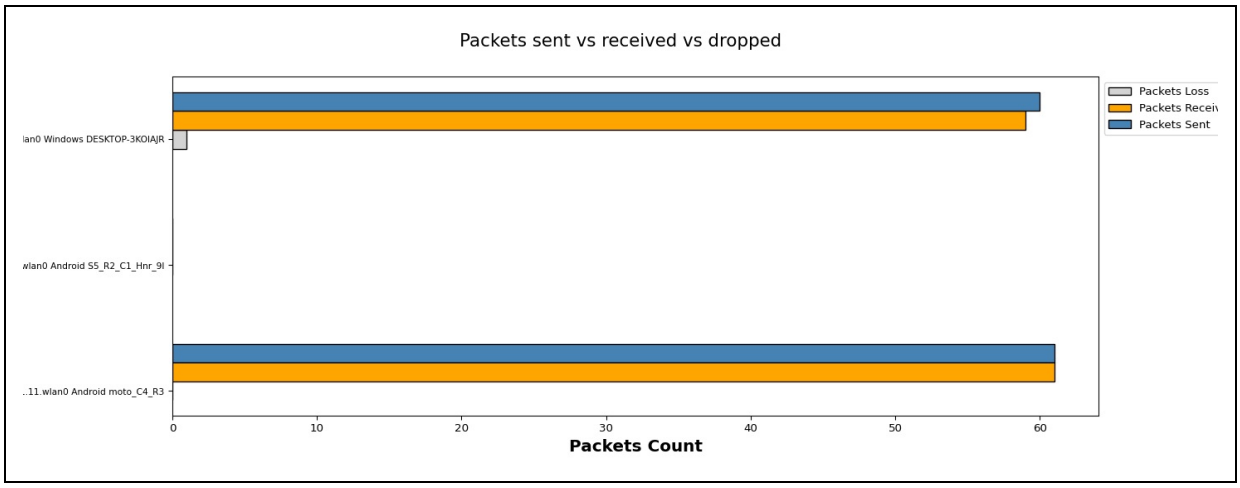
Sno	Test Cases	Test Duration	Test Status
1	Ping Test	1.0 minute	Executed
2	Quality Of Service(QOS) Test ['VO', 'VI', 'BE', 'BK']	1.0 minute	Executed
3	FTP Test	1.0 minute	Executed
4	HTTP Test	1.0 minute	Executed
5	Multicast Test	1.0 minute	Executed

1. Ping Test

Test Configuration

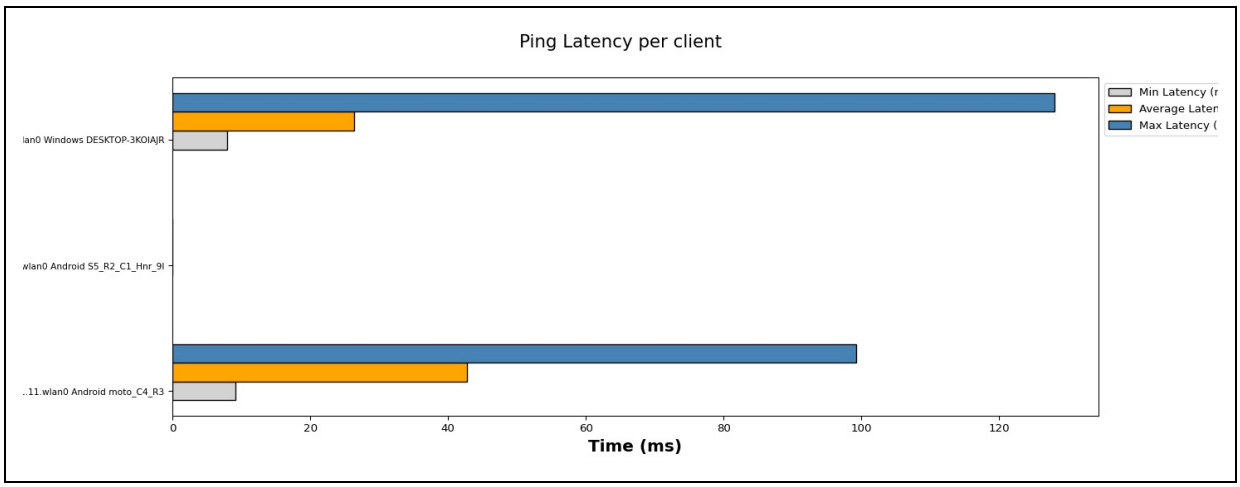
Test Setup Information	IP / Website	google.com

Packets sent vs packets received vs packets dropped



Wireless Client	MAC	Channel	SSID	Mode	Packets Sent	Packets Received	Packets Loss
moto_C4_R3 Android	0a:5b:89:56:a9:ae	36	OpenWifi_psk2	802.11abgn-AC 80	61	61	0
S5_R2_C1_Hnr_9I Android	c4:9f:4c:fc:9b:f3	0		AUTO 20	0	0	0
DESKTOP-3KOIAJR Windows	f4:8c:50:e2:01:67	36	OpenWifi_psk2	802.11abgn-AC 20 1x1	60	59	1

Ping Latency Graph



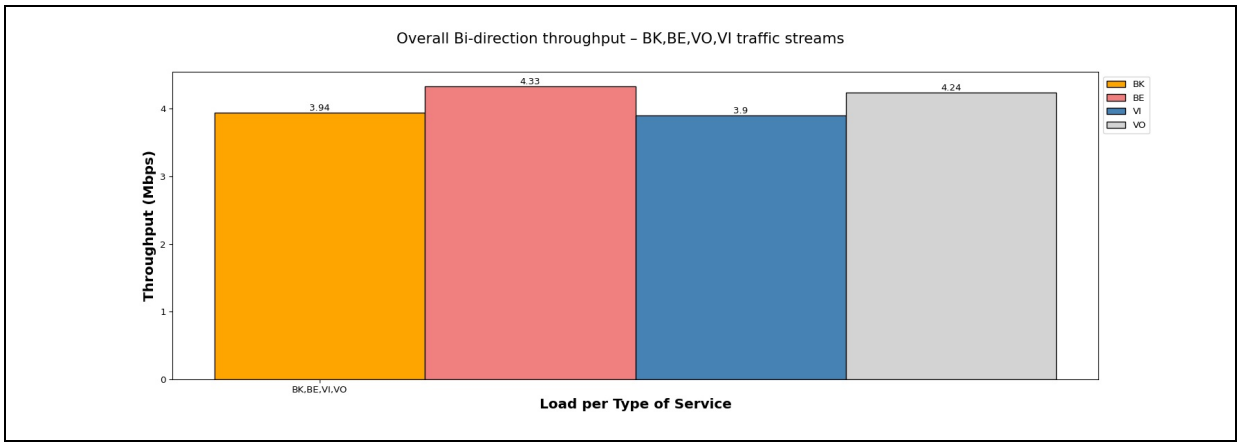
Wireless Client	MAC	Channel	SSID	Mode	Min Latency (ms)	Average Latency (ms)	Max Latency (ms)
moto_C4_R3 Android	0a:5b:89:56:a9:ae	36	OpenWifi_psk2	802.11abgn-AC 80	9.1	42.756	99.2
S5_R2_C1_Hnr_9I Android	c4:9f:4c:fc:9b:f3	0		AUTO 20	0.0	0.000	0.0
DESKTOP-3KOIAJR Windows	f4:8c:50:e2:01:67	36	OpenWifi_psk2	802.11abgn-AC 20 1x1	8.0	26.373	128.0

2. Quality Of Service(QOS) Test

Test Configuration

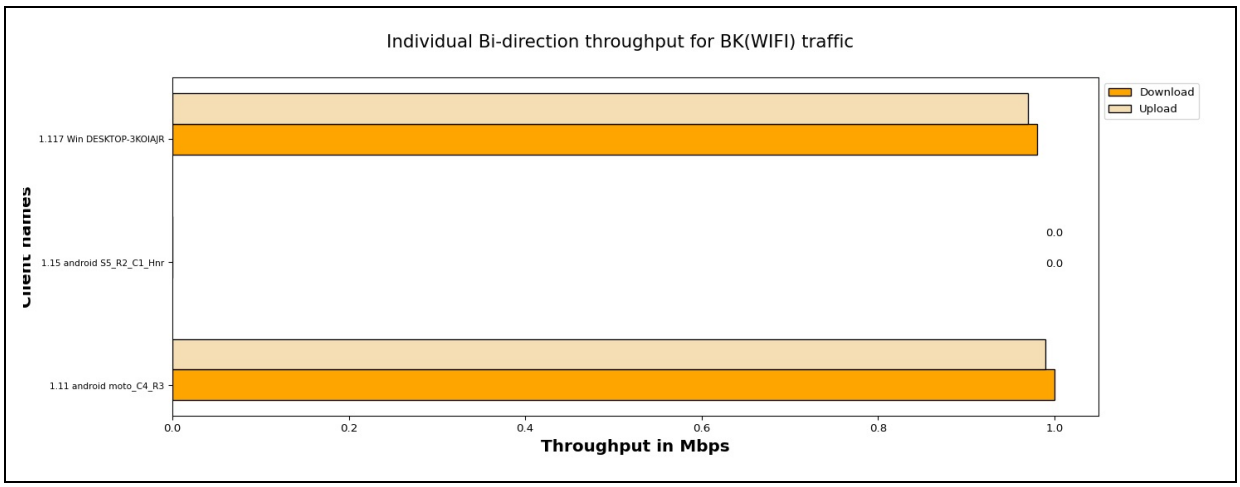
Test Setup Information	Protocol	TCP
	Traffic Direction	Bi-direction
	Security	Test Configured
	TOS	['VO', 'VI', 'BE', 'BK']
	Per TOS Load	Upload:1.0 Mbps,Download:1.0 Mbps

Overall Bi-direction Throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)



Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BK(WiFi).

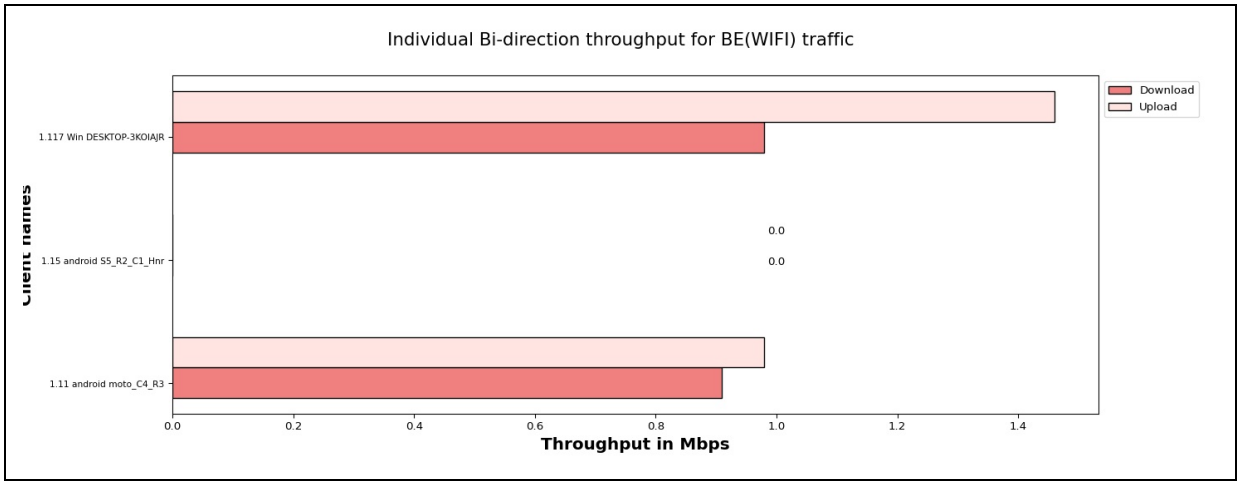
The below graph represents individual throughput for 3 clients running BK (WiFi) traffic. X- axis shows “Throughput in Mbps” and Y-axis shows “number of clients”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android moto_C4_R3	0a:5b:89:56:a9:ae	OpenWifi_psk2	Background	1.0 Mbps	1.0 Mbps	1.0 Mbps	0.99 Mbps	0.06	0.00
1.15 android S5_R2_C1_Hnr	c4:9f:4c:fc:9b:f3		Background	1.0 Mbps	1.0 Mbps	0.0 Mbps	0.0 Mbps	0.00	0.00
1.117 Win DESKTOP-3KOIAJR	f4:8c:50:e2:01:67	OpenWifi_psk2	Background	1.0 Mbps	1.0 Mbps	0.95 Mbps	0.96 Mbps	0.00	0.05

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BE(WiFi).

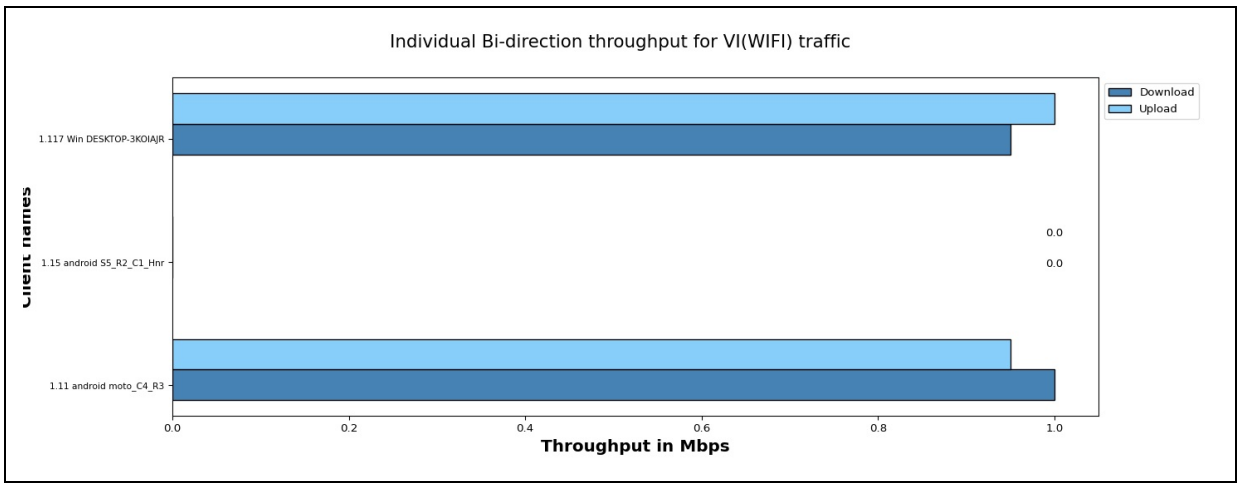
The below graph represents individual throughput for 3 clients running BE (WiFi) traffic. X- axis shows “number of clients” and Y-axis shows “Throughput in Mbps”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android moto_C4_R3	0a:5b:89:56:a9:ae	OpenWifi_psk2	Besteffort	1.0 Mbps	1.0 Mbps	1.01 Mbps	0.99 Mbps	0.0	0.00
1.15 android S5_R2_C1_Hnr	c4:9f:4c:fc:9b:f3		Besteffort	1.0 Mbps	1.0 Mbps	0.0 Mbps	0.0 Mbps	0.0	0.00
1.117 Win DESKTOP-3KOIAJR	f4:8c:50:e2:01:67	OpenWifi_psk2	Besteffort	1.0 Mbps	1.0 Mbps	1.0 Mbps	0.99 Mbps	0.0	0.01

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VI(WiFi).

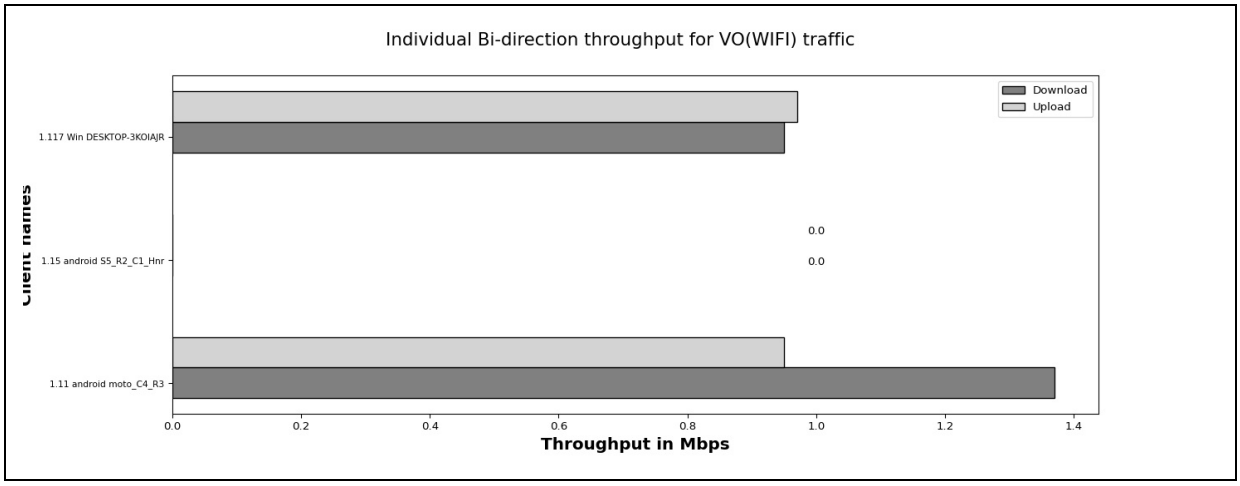
The below graph represents individual throughput for 3 clients running VI (WiFi) traffic. X- axis shows “number of clients” and Y-axis shows “Throughput in Mbps”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android moto_C4_R3	0a:5b:89:56:a9:ae	OpenWifi_psk2	Video	1.0 Mbps	1.0 Mbps	0.75 Mbps	0.75 Mbps	0.05	0.0
1.15 android S5_R2_C1_Hnr	c4:9f:4c:fc:9b:f3		Video	1.0 Mbps	1.0 Mbps	0.0 Mbps	0.0 Mbps	0.00	0.0
1.117 Win DESKTOP-3KOIAJR	f4:8c:50:e2:01:67	OpenWifi_psk2	Video	1.0 Mbps	1.0 Mbps	0.87 Mbps	0.86 Mbps	0.02	0.0

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 3 clients running VO (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android moto_C4_R3	0a:5b:89:56:a9:ae	OpenWifi_psk2	Voice	1.0 Mbps	1.0 Mbps	1.0 Mbps	1.0 Mbps	0.00	0.00
1.15 android S5_R2_C1_Hnr	c4:9f:4c:fc:9b:f3		Voice	1.0 Mbps	1.0 Mbps	0.0 Mbps	0.0 Mbps	0.00	0.00
1.117 Win DESKTOP-3KOIAJR	f4:8c:50:e2:01:67	OpenWifi_psk2	Voice	1.0 Mbps	1.0 Mbps	0.79 Mbps	0.78 Mbps	0.03	0.05

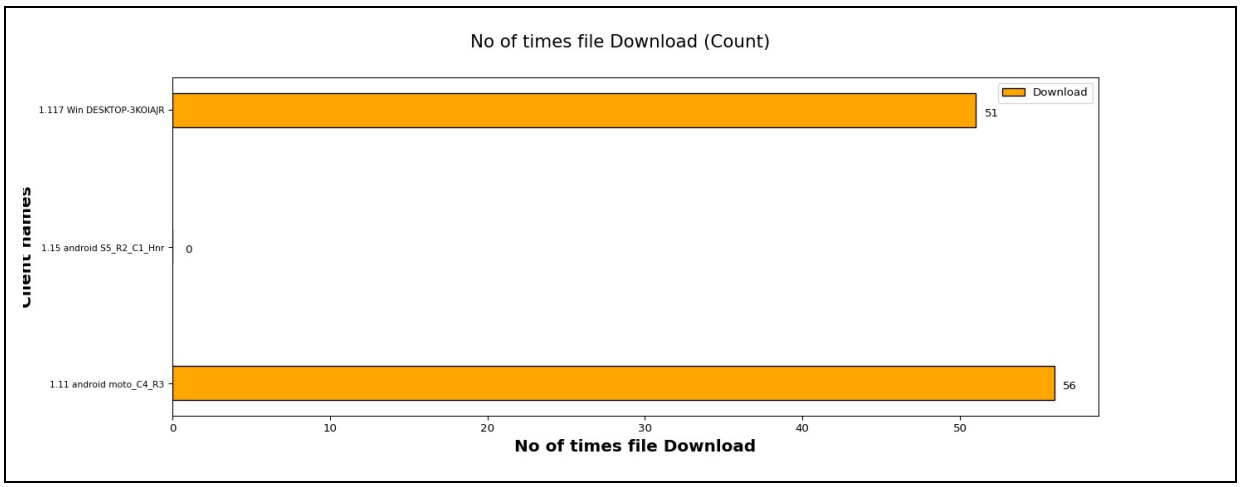
3. File Transfer Protocol (FTP) Test

Test Configuration

Test Setup Information	Traffic Direction	Download
	File Size	10MB
	File Location	/home/lanforge

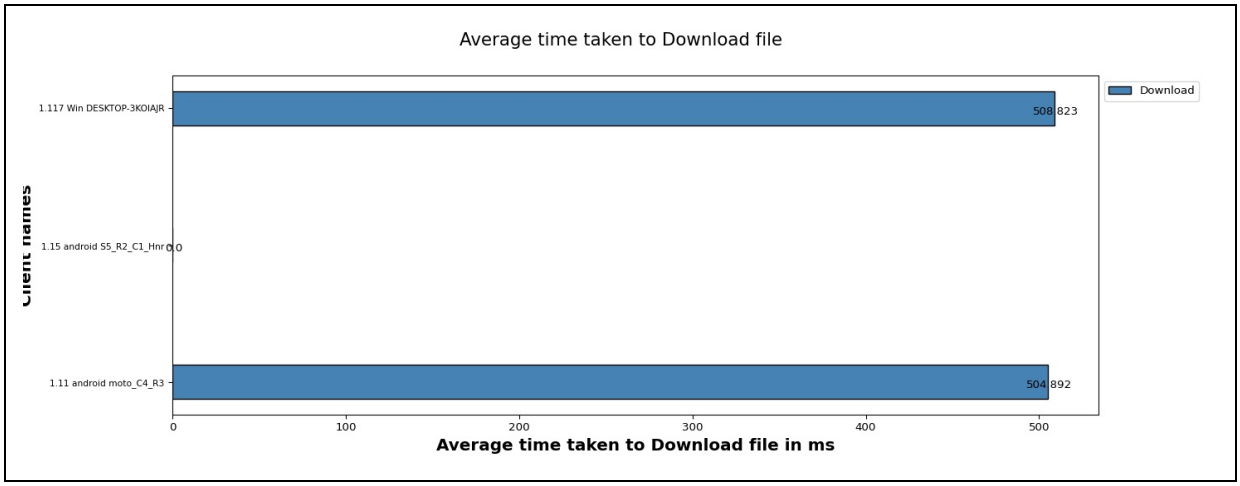
No.of times file Download

The below graph represents number of times a file Download for each client (WiFi) traffic. X-axis shows "No of times file Download" and Y-axis shows "Client names".



Average time taken to Download file

The below graph represents average time taken to Download for each client (WiFi) traffic. X-axis shows "Average time taken to Download a file " and Y-axis shows "Client names".



Overall Results

Clients	MAC	Channel	SSID	Mode	No of times File downloaded	Time Taken to Download file (ms)	Bytes-rd (Mega Bytes)	RX RATE (Mbps)
1.11 android moto_C4_R3	0a:5b:89:56:a9:ae	36	OpenWifi_psk2	802.11abgn-AC 80	56	504.892	560.0000	81.2176
1.15 android S5_R2_C1_Hnr	c4:9f:4c:fc:9b:f3	0		AUTO 20	0	0.000	0.0000	0.0000
1.117 Win DESKTOP-3KOIAJR	f4:8c:50:e2:01:67	36	OpenWifi_psk2	802.11abgn-AC 20 1x1	51	508.823	510.0146	77.7508

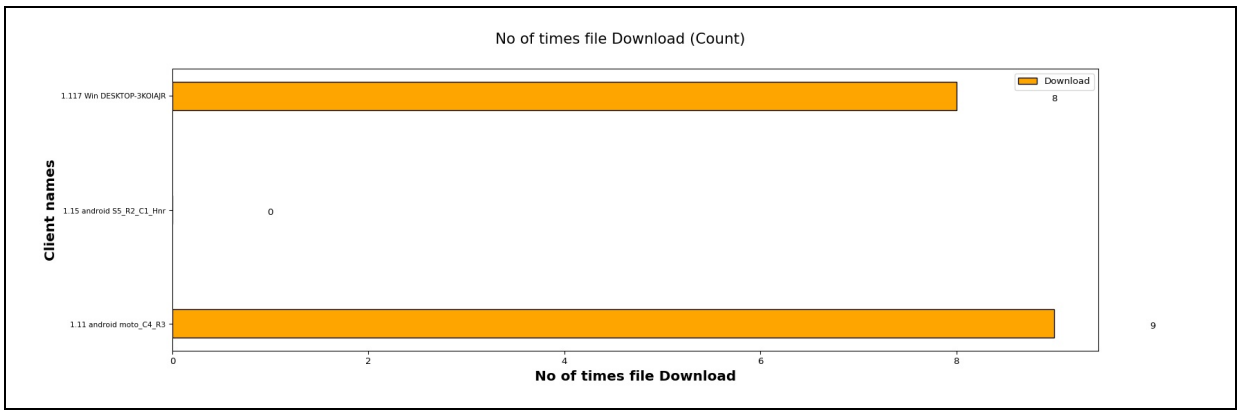
4. Hyper Text Transfer Protocol (HTTP) Test

Test Configuration

Test Setup Information	Traffic Direction	Download
	File Size	5MB
	File location	/usr/local/lanforge/nginx/html

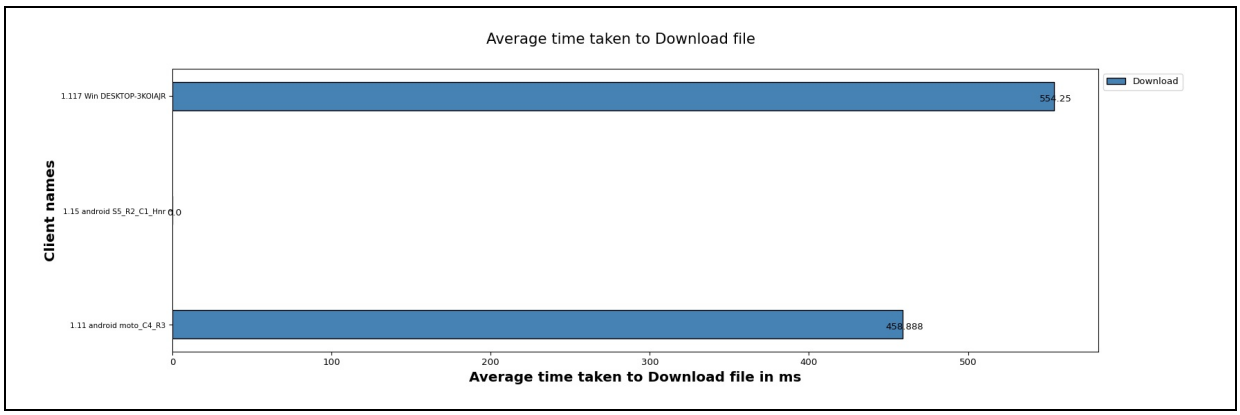
No of times file Downloads

The below graph represents number of times a file downloads for each client. X- axis shows "No of times file downloads and Y-axis shows Client names.



Average time taken to download file

The below graph represents average time taken to download for each client . X- axis shows "Average time taken to download a file " and Y-axis shows Client names.



Overall Results

Clients	MAC	Channel	SSID	Mode	No of times File downloaded	Average time taken to Download file (ms)	Bytes-rd (Mega Bytes)	Rx Rate (Mbps)
1.11 android moto_C4_R3	0a:5b:89:56:a9:ae	36	OpenWifi_psk2	802.11abgn-AC 80	9	458.888	45.0	7.3911
1.15 android S5_R2_C1_Hnr	c4:9f:4c:fc:9b:f3	0		AUTO 20	0	0.000	0.0	0.0000
1.117 Win DESKTOP-3KOIAJR	f4:8c:50:e2:01:67	36	OpenWifi_psk2	802.11abgn-AC 20 1x1	8	554.250	40.0	5.9088

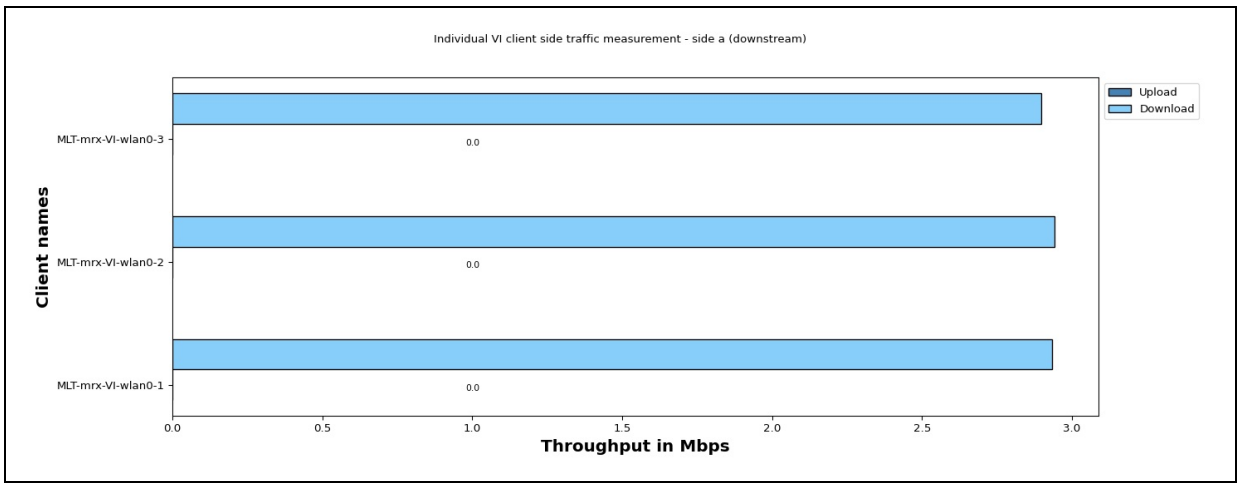
5. Multicast Test

Test Configuration

Test Setup Information	Protocol	UDP
	Type of Service (TOS)	VI
	Upstream Port	1.1.eth2
	Offered Load (Mbps)	3.0

Individual throughput with intended load 3.0 Mbps station for traffic VI (WiFi).

The below graph represents individual throughput for 3 clients running VI (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".



Client Name	Endp Name	HW Version	Port Name	Mode	Mac	SSID	Channel	Type of traffic	Traffic Protocol	Offered Download Rate Per Client (Mbps)	Download Rate Per Client (Mbps)	Download Drop Percentage (%)
1.11_ilclbld189_Android	MLT-mrx-VI-wlan0-1	motorola motorola one fusion+ r	1.11.wlan0	802.11abgn-AC 80	0a:5b:89:56:a9:ae	OpenWifi_psk2	36	VI	Mcast	3.0	2.934726	0.0
1.15_cn-east-1a-23a86a96615918984371_Android	MLT-mrx-VI-wlan0-2	HUAWEI LLD-AL10 r9 sdk: 28	1.15.wlan0	AUTO 20	c4:9f:4c:fc:9b:f3		0	VI	Mcast	3.0	2.941520	0.0
1.117_DESKTOP-3KOIAJR_Win	MLT-mrx-VI-wlan0-3	Win/x86 6.2	1.117.wlan0	802.11abgn-AC 20 1x1	f4:8c:50:e2:01:67	OpenWifi_psk2	36	VI	Mcast	3.0	2.898306	0.0

Overall Info	contact	support@candelatech.com